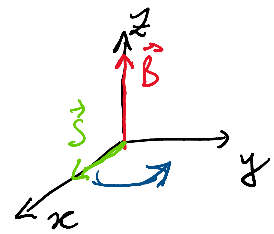
Esercizio spin precession

venerdì 8 ottobre 2021 08:36

$$\frac{\partial H}{\partial t} = 0 \qquad |\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$$



$$H = \frac{\Delta}{2} Z = \begin{pmatrix} \Delta/2 & 0 \\ 0 & -\Delta/2 \end{pmatrix} \qquad |\psi(0)\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}$$

$(\Delta \propto B)$

$$H|0\rangle = \frac{\Delta}{2}|0\rangle \qquad e^{-iHt/\hbar}|0\rangle = e^{-i\frac{\Delta t}{2\hbar}}|0\rangle$$

$$H|1\rangle = -\frac{\Delta}{2}|1\rangle \qquad e^{-iHt/\hbar}|1\rangle = e^{i\Delta t/2\hbar}|1\rangle$$

$$|\psi(t)\rangle = e^{-iHt/\hbar}\frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{e^{-i\Delta t/2\hbar}|0\rangle + e^{i\Delta t/2\hbar}|1\rangle}{\sqrt{2}} \qquad \frac{\Delta}{\hbar} = \omega$$

$$S = \frac{\hbar}{2} \qquad \hat{S}_z = \frac{\hat{Z}}{2}\hbar \qquad \hat{S}_z|0\rangle = \frac{\hbar}{2}|0\rangle \qquad \hat{S}_z|1\rangle = -\frac{\hbar}{2}|1\rangle$$

$$\langle \hat{S}_z \rangle = \langle \psi(t)|\hat{S}_z|\psi(t)\rangle = \qquad \langle \psi(t)| = \frac{1}{\sqrt{2}}\left(e^{i\omega t/2}\langle 0| + e^{-i\omega t/2}\langle 1|\right)$$

$$= \frac{\hbar}{4}\left(e^{i\omega t/2}\langle 0| + e^{-i\omega t/2}\langle 1|\right) \cdot \qquad \hat{S}_z = \frac{\hbar}{2}\left(|0\rangle\langle 0| - |1\rangle\langle 1|\right) = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\left(|0\rangle\langle 0| - |1\rangle\langle 1|\right) \cdot \left(e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle\right) \qquad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \frac{\hbar}{4}\left(e^{i\omega t/2}\langle 0|0\rangle\langle 0|0\rangle e^{-i\omega t/2} - e^{-i\omega t/2}\langle 1|1\rangle\langle 1|1\rangle e^{i\omega t/2}\right) =$$

$$= \frac{\hbar}{4}(1 - 1) = 0$$

$$\hat{S}_x = \frac{\hbar}{2}X = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2}\left(|0\rangle\langle 1| + |1\rangle\langle 0|\right)$$

$$\langle \hat{S}_x \rangle = \langle \psi(t)|\hat{S}_x|\psi(t)\rangle =$$

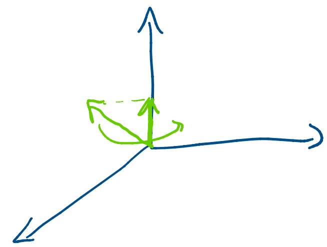
$$= \frac{\hbar}{4}\left(e^{i\omega t/2}\langle 0| + e^{-i\omega t/2}\langle 1|\right)$$

$$\left(|0\rangle\langle 1| + |1\rangle\langle 0|\right)\left(e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle\right) =$$

$$\left(\; e^{i\omega t} \qquad e^{-i\omega t} \; \right)$$

$$= \frac{\hbar}{4}\left(e^{i\omega t}\langle 0|0\rangle\langle 1|1\rangle + e^{-i\omega t}\langle 1|1\rangle\langle 0|0\rangle\right) =$$

$$= \frac{\hbar}{2}\left(\frac{e^{i\omega t}+e^{-i\omega t}}{2}\right) = \frac{\hbar}{2}\cos\omega t$$



$[\hat{S}_z, \hat{H}] = 0$ $\quad \frac{\partial S_z}{\partial t} = 0 \Rightarrow S_z$ cost. del moto

$[\hat{S}_x, \hat{H}] \neq 0$ $\quad$ varia

$$\hat{S}_y = \frac{\hbar}{2} Y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar}{2}\left(-i|0\rangle\langle 1| + i|1\rangle\langle 0|\right)$$

$$Y|0\rangle = i|1\rangle$$
$$Y|1\rangle = -i|0\rangle$$

$$\langle \hat{S}_y \rangle = \frac{\hbar}{2}\sin\omega t$$